

Module/Pathway/Taxon Review —

Methylglyoxal Detoxification in

Pseudomonas putida KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Nominal bucket:** KEGG `ppu00620` "Pyruvate metabolism" (module area: central carbon) **Review focus:** methylglyoxal (MG) detoxification module satisfiability and gene-annotation curation

1. Executive Summary

The methylglyoxal (MG) detoxification module is **fully satisfiable** in *Pseudomonas putida* KT2440 through a complete, canonical, glutathione (GSH)-dependent glyoxalase system. Both enzymatic steps are encoded: **glyoxalase I** (`gloA` = PP_3766, lactoylglutathione lyase, EC 4.4.1.5) and **glyoxalase II** (`gloB` = PP_4144, hydroxyacylglutathione hydrolase, EC 3.1.2.6). These two enzymes sequentially convert the spontaneously-formed GSH–MG hemithioacetal into D-lactate and regenerate GSH. Crucially, the *P. putida* glyoxalase I gene has **direct experimental validation**: heterologous expression detoxified MG in *E. coli* and in mammalian (CHO) cells, giving PP_3766 an unusually strong evidence footing for a KT2440 metabolic gene. The GSH cofactor required by this route is itself synthesizable, since both de novo biosynthesis steps — `gshA` (PP_0243) and `gshB` (PP_4993) — are encoded, so the module is cofactor-satisfiable and does not depend on GSH import.

The one "step" that is **not expected** in this organism is the *generation* of MG by a dedicated methylglyoxal synthase. Exhaustive UniProt searches of the KT2440 proteome (and of all *P. putida*) for methylglyoxal synthase / `mgsA` / EC 4.2.3.3 returned **zero hits**. In KT2440, MG arises non-enzymatically (and as a metabolic side-product) rather than through the *E. coli*-style MgsA bypass. This is a meaningful curation distinction: a generic MG module that expects an MG synthase should mark that step `not_expected_in_target_taxon` for KT2440 rather than as a gap.

Finally, this review flags a mismatch between the delivered "pyruvate metabolism" candidate list (54 genes, dominated by central-carbon enzymes) and the biology of the MG-detox module. The genuinely MG-relevant genes are a small subset (`gloA`, `gloB`, plus downstream lactate dehydrogenases feeding D-lactate to pyruvate), while several real alternative-route detox candidates are **absent from the candidate metadata** and should be curated separately: a second glyoxalase-II-family paralog (**PP_0772**), reductive aldo-keto reductases (`yeaE` / **PP_3120**, `dkgB` / **PP_2368**), and DJ-1/ThiJ/PfpI-superfamily glyoxalase-III candidates (**PP_0893**, **PP_2491**, **PP_2992**, **PP_3431**). Conversely, `ycgM` (PP_5153), despite sitting in the `ppu00620` bucket, is a fumarylacetoacetase-family enzyme unrelated to MG and should be excluded from the module.

2. Target-Organism Pathway Definition

Process included. "Methylglyoxal detoxification" is the conversion of the reactive dicarbonyl 2-oxopropanal (methylglyoxal, MG) — a cytotoxic electrophile that glycates DNA, RNA, and protein — into a non-toxic hydroxy acid. In the canonical glutathione-dependent route the module comprises exactly two catalytic steps operating on the non-enzymatically formed GSH–MG adduct:

1. **Glyoxalase I (GlxI, EC 4.4.1.5):** isomerizes the hemithioacetal of GSH + MG to (R)-S-lactoylglutathione.
2. **Glyoxalase II (GlxII, EC 3.1.2.6):** hydrolyzes S-lactoylglutathione to **D-lactate**, regenerating GSH.

The D-lactate product then enters central carbon metabolism (oxidation to pyruvate), which is why the module is administratively bucketed under KEGG **ppu00620 "Pyruvate metabolism."**

Boundaries — neighboring processes to keep separate. - **MG generation** (methylglyoxal synthase, `mgsA`, EC 4.2.3.3, from dihydroxyacetone phosphate) is a distinct, catabolic-overflow process. It is a separate pathway node and, as shown below, is not encoded in KT2440. - **Glutathione metabolism (ppu00480):** GSH biosynthesis (`gshA` / `gshB`) and recycling is an upstream cofactor-supply module, not part of the two-step detox module itself, but it gates satisfiability. - **Downstream lactate oxidation** (D-/L-lactate dehydrogenases; pyruvate metabolism proper) is a separate, well-populated central-carbon module. - **Glycolysis / triose-phosphate handling (ppu00010):** the non-enzymatic *source* of MG, not its detox. - **GSH-**

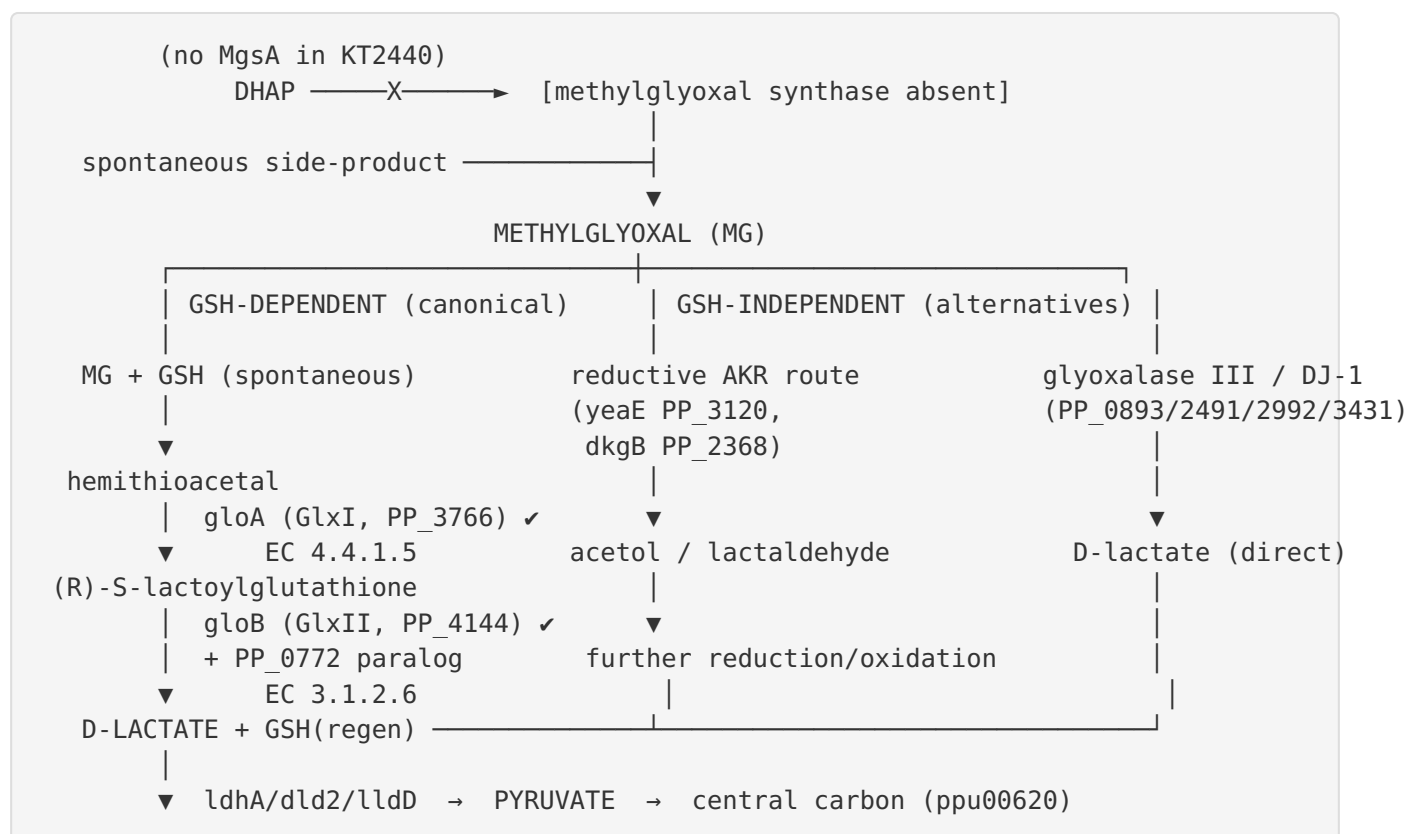
independent MG detox (aldo-keto reductase reductive route to acetol/lactaldehyde; DJ-1/glyoxalase-III direct hydrolysis to lactate) are alternative modules/routes that should be documented separately, not merged into the glyoxalase module.

Alternate names / database definitions. Glyoxalase system; GlxI / GloA / lactoylglutathione lyase / aldoketomutase (EC 4.4.1.5); GlxII / GloB / hydroxyacylglutathione hydrolase / S-2-hydroxyacylglutathione hydrolase (EC 3.1.2.6); "methylglyoxal degradation" (MetaCyc/BioCyc). The broad KEGG "Pyruvate metabolism" map (ppu00620) is an overview bucket and is *not* an MG-specific module — hence the candidate list is dominated by unrelated central-carbon enzymes.

Cofactor requirement (verified). The glyoxalase route depends on a regenerated glutathione pool; KT2440 encodes complete de novo GSH biosynthesis — `gshA` (PP_0243, EC 6.3.2.2) and `gshB` (PP_4993, EC 6.3.2.3), plus a second putative γ -glutamylcysteine ligase (PP_3253) — so the primary route is satisfiable at both the enzyme and cofactor level.

3. Expected Step Model

Step	Reaction	Enzyme / EC	Expected in KT2440?	Encoded by
(0) MG generation	DHAP → methylglyoxal + Pi	Methylglyoxal synthase, EC 4.2.3.3 (mgsA)	No (not_expected_in_target_taxon)	<i>none found</i>
(pre) Adduct formation	MG + GSH → hemithioacetal	non-enzymatic	Yes (spontaneous)	—
1	hemithioacetal → (R)-S-lactoylglutathione	Glyoxalase I, EC 4.4.1.5	Yes (covered)	gloA / PP_3766
2	S-lactoylglutathione + H ₂ O → D-lactate + GSH	Glyoxalase II, EC 3.1.2.6	Yes (covered)	gloB / PP_4144 (+ paralog PP_0772)
cofactor	Glu+Cys→γ-EC→GSH	GshA EC 6.3.2.2 / GshB EC 6.3.2.3	Yes (covered)	PP_0243 / PP_4993
downstream	D-lactate → pyruvate	D-/L-lactate DH	Yes (covered)	ldhA / PP_1649, dld2 / PP_4737, lldD / PP_4736
alt route A	MG → acetol / lactaldehyde (reductive, GSH-independent)	Aldo-keto reductase, EC 1.1.1.-	Likely (candidate_uncertain)	yeaE / PP_3120, dkgB / PP_2368
alt route B	MG → D-lactate directly (GSH-independent)	Glyoxalase III / DJ-1, EC 4.2.1.130	Possible (candidate_uncertain)	PP_0893, PP_2491, PP_2992, PP_3431



4. Candidate Genes and Evidence

4.1 High-confidence, module-defining genes

gloA — PP_3766 (Q88GF8) — Glyoxalase I (EC 4.4.1.5). This is the core, rate-relevant enzyme of the module. UniProt annotates it as lactoylglutathione lyase with the catalytic activity (R)-S-lactoylglutathione = methylglyoxal + glutathione, and it carries Ni/Zn metal-binding keywords consistent with a metalloisomerase glyoxalase I. It maps to KEGG ppu00620. Its UniProt existence level is 3 (inferred from homology), but the *P. putida* glyoxalase I gene itself has **direct functional evidence** independent of that annotation: heterologous expression in *E. coli* and in Chinese hamster ovary cells produced substantial MG reductions (35–43% intracellular MG in *E. coli*; 80–90% in CHO cells) (PMID: 11461144; PMID: 22358913). *P. putida* GlxI is a classically studied enzyme, and its metal-activation class (Ni²⁺ vs Zn²⁺) and distinctive loop/helix features have been characterized (PMID: 10824093; PMID: 25557363). **Curation call:**

covered (high confidence). Promote to full `fetch-gene` review to attach the experimental citations, lock metal class, and confirm the KT2440 locus corresponds to the historically studied *P. putida* GlxI.

gloB — **PP_4144 (Q88FF3)** — **Glyoxalase II (EC 3.1.2.6)**. Annotated as hydroxyacylglutathione hydrolase (S-(2-hydroxyacyl)glutathione + H₂O = 2-hydroxy carboxylate + glutathione), mapping to ppu00620, existence level 3. InterPro signatures are specific to the GlxII subfamily: IPR017782 (hydroxyacylglutathione hydrolase), IPR035680 (Glx_II metallo-β-lactamase), and an HAGH_C domain — i.e., a bona fide glyoxalase II, not merely a generic metallo-β-lactamase. **Curation call: covered.** One important caveat for module *weighting*: in *E. coli*, a ΔgloB mutant remains as MG-tolerant as wild type despite showing the same block in MG detoxification as ΔgloA ([PMID: 21143325](#)). This indicates GlxII is not the tolerance-limiting step — the accumulation and disposal of S-lactoylglutathione at the GlxI step is what governs tolerance. For satisfiability the module still needs a GlxII activity, but curators should not treat **gloB** as the essential determinant.

gshA — **PP_0243 (Q88R90)** — **glutamate-cysteine ligase (EC 6.3.2.2)** and **gshB** — **PP_4993 (Q88D35)** — **glutathione synthetase (EC 6.3.2.3)**. Both de novo GSH-biosynthesis steps are encoded (a second putative γ-glutamylcysteine ligase, PP_3253, also exists), so the GSH cofactor consumed transiently and regenerated by the glyoxalase cycle is supplied endogenously. **Curation call: cofactor supply covered.** These belong to glutathione metabolism, not the detox module proper, but they gate its satisfiability and should be cross-linked.

4.2 Downstream disposal (genuinely in ppu00620)

The D-lactate produced by GlxII is returned to pyruvate metabolism by lactate dehydrogenases already in the candidate list: **ldhA**/PP_1649 (D-lactate DH, EC 1.1.1.28), **dld2**/PP_4737 (D-lactate DH, EC 1.1.-.-), and **lldD**/PP_4736 (L-lactate DH). These are correctly bucketed and give the module a plausible sink into central carbon. **Curation call: covered (downstream), low ambiguity** — though these are multifunctional central-carbon enzymes and should not be labeled MG-specific.

4.3 Second glyoxalase-II candidate (present in bucket, missing from list logic)

PP_0772 (Q88PS6) — **"Metallo-β-lactamase family protein," ppu00620**. InterPro classifies PP_0772 with IPR051453 "MBL_Glyoxalase_II" together with IPR001279 (metallo-β-lactamase fold) and IPR036866 (RNase Z / hydroxyglutathione hydrolase). It therefore falls in the **glyoxalase-II subfamily** and is a credible second GlxII-family activity, distinct from the

primary **gloB** (which carries the more specific IPR017782/IPR035680/HAGH_C signatures). Note that the IPR051453 subfamily also contains non-glyoxalase metallo-hydrolases (persulfide dioxygenase/ETHE1, tRNase Z), so specificity is not guaranteed. **Curation call: candidate_uncertain; promote to full review** to resolve whether PP_0772 is a true GlxII, an ETHE1-type sulfur dioxygenase, or a tRNase Z.

4.4 Genes in the bucket that are NOT MG-related

ycgM — PP_5153 (Q88CM7) — "Isomerase/hydrolase (EC 3.-.-.)," **ppu00620**. Despite the vague EC and its placement in the pyruvate bucket, InterPro assigns it to the **fumarylacetoacetase-like (FAH) family** (IPR011234 / PF01557), a hydratase/decarboxylase fold with no glyoxalase activity. **Curation call: exclude from the MG module** (over-broad EC placeholder driving spurious inclusion).

The remaining ~48 candidate genes (PDH complex **aceE** / **aceF** / **lpd**, ACC **accABCD**, malate/fumarate enzymes, pyruvate kinases, PEP carboxylase/synthase, acetyl-CoA synthetases, thiolases, quinoprotein/alcohol dehydrogenases, etc.) are legitimate pyruvate/central-carbon enzymes but have **no role in MG detoxification** and should not be scored against this module.

4.5 Alternative-route detox genes MISSING from the candidate metadata

These are real KT2440 proteome members not present in the delivered ppu00620 candidate list, and they should be added as separate MG-related candidates:

Gene / locus	UniProt	Annotation	Proposed MG role	Confidence
yeaE / PP_3120	Q88I81	Methylglyoxal reductase (EC 1.1.1.-), AKR (PF00248)	Reductive, GSH- independent MG→acetol/ lactaldehyde	Moderate (name + fold)
dkgB / PP_2368	Q88KC2	Aldo-keto reductase; hydroxyacetone + NADP ⁺ = MG + NADPH	Reductive MG↔acetol	Moderate
PP_0893, PP_2491, PP_2992, PP_3431	—	DJ-1/ThiJ/PfpI- superfamily	Candidate glyoxalase III (GSH-independent MG→D-lactate)	Low-moderate (homology only)

The DJ-1/ThiJ flag is justified by demonstrated glutathione-independent glyoxalase activity in DJ-1-superfamily proteins in other organisms (e.g., *Candida albicans* GLX3) (PMID: 24302734). Transfer to KT2440 is by homology only and is **uncertain** — the same fold also yields proteases (e.g., PfpI-type), a classic mis-annotation trap — so these are worth flagging so the module is not scored as a single point of failure on `gloA`.

5. Mechanistic Interpretation

The satisfiability logic of this module is best understood as a **core route plus optional parallel branches**:

- **Core (covered):** MG spontaneously forms a hemithioacetal with GSH; **GlxI** (`gloA`/PP_3766) isomerizes it and **GlxII** (`gloB`/PP_4144) hydrolyzes the product to D-lactate, recycling GSH. Because GSH is regenerated stoichiometrically and is itself synthesizable (`gshA` / `gshB`), the route is self-sustaining and requires no external cofactor input. Direct heterologous functional data for *P. putida* GlxI make this the best-evidenced branch.
 - **Rate/tolerance logic:** GlxI is the committed, tolerance-governing step; GlxII is required to complete turnover and recycle GSH but is not, by itself, the determinant of MG resistance (the *E. coli* Δ gloB phenotype). Curators should therefore weight `gloA` presence more heavily than `gloB` when scoring "detox capability," while still requiring both for a complete pathway.
 - **Parallel branches (candidate_uncertain):** KT2440 additionally carries reductive (AKR: `yeaE`, `dkgB`) and potentially GSH-independent (DJ-1/ThiJ glyoxalase III) routes. These provide functional redundancy — biologically plausible for a soil bacterium that experiences oxidative and carbonyl stress — but are not needed to declare the module satisfiable, and their activity in KT2440 is unproven.
 - **Boundary correction:** The absence of an MG synthase means the module in KT2440 is purely *defensive* (detoxifying incidentally-formed MG), not a deliberate glycolytic-bypass sink as in some enterobacteria. This reframes the upstream step from "gap" to "not expected."
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6. Evidence Base

PMID	Finding it supports	Organism / transfer strength
11461144	<i>P. putida</i> glyoxalase I detoxifies MG (35–43% intracellular MG reduction) when expressed in <i>E. coli</i> — direct functional support for <code>gloA</code> /PP_3766	<i>P. putida</i> gene, heterologous host — strong transfer to KT2440
22358913	<i>P. putida</i> GlxI expressed in CHO cells lowers free MG 80–90% — corroborates <code>gloA</code> function	<i>P. putida</i> gene — strong
28939611	Defines the GloA/GloB GSH-dependent MG→D-lactate two-enzyme system	Generic bacterial framework — strong conceptual
21143325	Δ gloB is as MG-tolerant as parent; S-lactoylglutathione formation is critical — GlxII non-essentiality caveat	<i>E. coli</i> — moderate (mechanism likely conserved)
24302734	DJ-1 superfamily member has GSH-independent glyoxalase (glyoxalase III) activity — justifies flagging KT2440 DJ-1/ThiJ paralogs	<i>C. albicans</i> — weak/uncertain transfer
10824093	Bacterial GlxI sequence identification; <i>P. putida</i> GlxI is distinctive among bacterial GlxI	Cross-genus — context only
25557363	Structural determinants of GlxI Ni ²⁺ vs Zn ²⁺ metal selectivity	<i>P. aeruginosa</i> — genus-level context

Verbatim supporting quotes recorded during the investigation: "*the transgenic cells with the P. putida glyoxalase I displayed a reduction of 35-43% in intracellular MG*"

([P 11461144](#));

"Glyoxalase I and II (GloA and GloB) sequentially convert MG into d-lactic acid using glutathione (GSH) as a cofactor"

([P 28939611](#));

"a Δ gloB mutant is as tolerant of MG as the parent, despite having the same degree of inhibition of MG detoxification as a Δ gloA strain"

([P 21143325](#));

"the DJ-1 superfamily member ORF 19.251/GLX3 from *Candida albicans* is shown to possess glyoxalase activity"

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P 24302734

7. Gaps, Ambiguities, and Likely Over-Annotations

- **Gap / boundary correction — MG synthase.** UniProt searches over UP000000556 for protein name "methylglyoxal synthase," gene `mgsA`, and EC 4.2.3.3 all returned zero hits; a broader search across all *P. putida* (taxonomy 303) for EC 4.2.3.3 was also empty. Free-text "methylglyoxal" in the KT2440 proteome returns only detox/reductase enzymes (`gloA`, `gloB`, `yeaE` PP_3120, `dkgB` PP_2368), never a synthase. Mark the synthesis step **not_expected_in_target_taxon** and document that MG arises non-enzymatically (triose-phosphate decomposition) and from aminoacetone/threonine and lipid-peroxidation routes. This is a negative database result and should be confirmed by manual synteny/HMM search.
- **Bucket/module mismatch (module_needs_revision at the bucket level).** The ppu00620 "pyruvate metabolism" bucket is an overview map, not an MG module. It (a) sweeps in ~48 unrelated central-carbon enzymes, (b) includes a non-glyoxalase (`ycgM`/PP_5153) apparently on the strength of a placeholder EC, and (c) omits the true alternative-route MG genes (PP_3120, PP_2368, DJ-1 paralogs). A dedicated **methylglyoxal-detoxification module document** would be far more curatable than scoring against ppu00620.
- **Over-propagation risk:** broad EC "1.1.1.-" and AKR annotations (`yeaE`, `dkgB`) may be over-mapped to MG; treat MG reductase activity as candidate, not established, for KT2440. `ycgM` EC 3.-.- is a meaningless placeholder driving mis-inclusion; correct family is fumarylacetoacetase.
- **Paralog ambiguity:** four DJ-1/ThiJ paralogs — cannot assign glyoxalase III to a specific locus by homology alone. PP_0772 vs `gloB` — two GlxII-family proteins; which carries the physiological S-lactoylglutathione hydrolase load in KT2440 is untested.
- **Evidence level:** all focal genes are UniProt existence level 3 (inferred from homology) *within KT2440*; the strongest experimental transfer is for `gloA` (the *P. putida* gene is functionally validated).

8. Module and GO-Curation Recommendations

Module element	Recommended status	Rationale
GlxI step (EC 4.4.1.5)	covered — <code>gloA</code> /PP_3766	Homology annotation + direct heterologous functional data for <i>P. putida</i> GlxI
GlxII step (EC 3.1.2.6)	covered — <code>gloB</code> /PP_4144 (paralog PP_0772)	Specific GlxII InterPro signatures; caveat: not tolerance-limiting
GSH cofactor supply	covered (cross-link) — <code>gshA</code> /PP_0243, <code>gshB</code> /PP_4993	Both biosynthesis steps encoded
D-lactate disposal	covered (downstream) — <code>ldhA</code> / <code>ldd2</code> / <code>lldD</code>	Multifunctional central-carbon LDHs
MG synthase (EC 4.2.3.3)	not_expected_in_target_taxon	Zero hits across KT2440 and all <i>P. putida</i>
Second GlxII paralog PP_0772	candidate_uncertain	GlxII subfamily fold; substrate untested
Reductive route (<code>yeaE</code> , <code>dkgB</code>)	candidate_uncertain (separate branch)	AKR fold + MG reductase naming; not in bucket
Glyoxalase III (DJ-1 paralogs)	candidate_uncertain (separate branch)	Homology only; activity unproven in KT2440
<code>ycgM</code> /PP_5153	exclude / module_needs_revision	Fumarylacetoacetase family, not glyoxalase
ppu00620 bucket as MG module	module_needs_revision	Overview map; create dedicated MG-detox module

GO-curation notes. Appropriate GO terms for the covered steps: GlxI → GO:0004462 (lactoylglutathione lyase activity) and GlxII → GO:0004416 (hydroxyacylglutathione hydrolase activity), both under GO:0051596 (methylglyoxal catabolic process) / GO:0019243 (methylglyoxal catabolic process to D-lactate via S-lactoylglutathione). A GO-term request is likely warranted only for the **glutathione-independent glyoxalase (glyoxalase III / DJ-1)**

branch (activity ~GO:0019172) if those paralogs are experimentally validated. No new module document is strictly required to mark satisfiability, but a dedicated MG-detox module would materially improve curation over the ppu00620 overview bucket.

9. Genes to Promote to Full fetch-gene Review

1. **gloA** / **PP_3766 (Q88GF8)** — module-defining GlxI; attach direct experimental citations
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confirm metal class (Ni vs Zn), and verify locus identity to historically characterized *P. putida* GlxI. **Highest priority.**
2. **gloB** / **PP_4144 (Q88FF3)** — module-defining GlxII; record the "not tolerance-limiting" caveat
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3. **PP_0772 (Q88PS6)** — second GlxII-family candidate; resolve whether it is a redundant S-lactoylglutathione hydrolase, an ETHE1-type persulfide dioxygenase, or a tRNase Z.
4. **yeaE** / **PP_3120 (Q88I81)** and **dkgB** / **PP_2368 (Q88KC2)** — reductive GSH-independent MG detox; confirm MG as physiological substrate and product (acetol vs lactaldehyde).
5. **DJ-1/ThiJ** paralogs **PP_0893**, **PP_2491**, **PP_2992**, **PP_3431** — glyoxalase III candidates; low prior but high value if positive.
6. (Cross-link only) **gshA** / **PP_0243** and **gshB** / **PP_4993** — cofactor supply.

10. Limitations and Knowledge Gaps

- **Evidence type is mixed.** The GlxI/GlxII annotations in KT2440 are UniProt existence level 3 (homology-inferred). The strongest experimental support

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is for "the *P. putida* glyoxalase I gene" used as a heterologous tool; it is very likely the PP_3766 ortholog but the exact KT2440 locus identity should be confirmed during full review. No KT2440-specific knockout/phenotype study of `gloA` / `gloB` was located in this investigation.

- **Alternative routes are homology-only.** The reductive AKR route and the DJ-1 glyoxalase-III branch are inferred from fold/family and from cross-species precedent (*C. albicans*, `P 24302734`).

Transfer to KT2440 is **uncertain** and untested biochemically.

- **Flux partitioning unknown.** How much MG is handled by the GSH-dependent vs GSH-independent routes under physiological/stress conditions in KT2440 has not been measured.
- **PP_0772 substrate scope unresolved** — metallo- β -lactamase-superfamily proteins are notoriously promiscuous, so InterPro placement in the GlxII subfamily is suggestive, not definitive.
- **Negative results are robust but bounded.** The absence of `mgsA` / EC 4.2.3.3 is well-supported across KT2440 and all *P. putida*, but "absent from UniProt-annotated proteome" cannot fully exclude an unannotated or divergent synthase; this is a low-probability caveat.

11. Proposed Follow-up Experiments / Actions

1. **Confirm PP_3766 $\leftrightarrow$ historical *P. putida* GlxI** by sequence alignment to the clones used in

`P 11461144`

22358913; this converts homology-level annotation to direct functional evidence for the KT2440 locus.

2. **Targeted deletion / complementation of `gloA` (PP_3766) and `gloB` (PP_4144)** in KT2440, scoring MG sensitivity and D-lactate accumulation; test the *E. coli*-derived prediction that Δ `gloB` is MG-tolerant while Δ `gloA` is sensitive.
3. **Biochemical assay of PP_0772** against S-lactoylglutathione to determine whether it is a redundant GlxII.
4. **Substrate assays for PP_3120 (`yeaE`) and PP_2368 (`dkgB`)** with MG $\pm$ NADPH to validate the reductive branch and its product.

5. **Express and assay the four DJ-1/ThiJ paralogs** for glutathione-independent MG→D-lactate (glyoxalase III) activity.
 6. **Curatorial action:** create a dedicated MG-detox module (GlxI + GlxII, GSH-supply cross-link, optional GSH-independent branches); remove `ycgM`/PP_5153; add PP_3120, PP_2368, and DJ-1 paralogs as candidate genes; mark the MG-synthase step `not_expected_in_target_taxon`.
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12. Key References

- *Accumulation of methylglyoxal in anaerobically grown Escherichia coli and its detoxification by expression of the Pseudomonas putida glyoxalase I gene.* PMID: 11461144 — direct functional support for PP_3766 (gloA).
- *Effect of endogenous methylglyoxal on Chinese hamster ovary cells grown in culture.* PMID: 22358913 — *P. putida* GlxI lowers free MG 80–90% in CHO cells.
- *Concomitant Loss of the Glyoxalase System and Glycolysis Makes "Candidatus Liberibacter asiaticus" an Energy Scavenger.* PMID: 28939611 — defines the GloA/GloB GSH-dependent system.
- *The critical role of S-lactoylglutathione formation during methylglyoxal detoxification in Escherichia coli.* PMID: 21143325 — GlxII non-essentiality caveat.
- *A glutathione-independent glyoxalase of the DJ-1 superfamily plays an important role in managing metabolically generated methylglyoxal in Candida albicans.* PMID: 24302734 — glyoxalase III precedent for DJ-1/ThiJ paralogs.
- *Identification of sequences encoding the detoxification metalloisomerase glyoxalase I in microbial genomes.* PMID: 10824093 — *P. putida* GlxI is distinctive among bacterial GlxI.
- *Modulating glyoxalase I metal selectivity by deletional mutagenesis.* PMID: 25557363 — Ni²⁺/Zn²⁺ GlxI activation classes (genus-level context).
- UniProt proteome UP000000556: gloA Q88GF8, gloB Q88FF3, PP_0772 Q88PS6, yeaE Q88I81, dkgB Q88KC2, gshA Q88R90, gshB Q88D35, DJ-1/ThiJ PP_0893/PP_2491/PP_2992/PP_3431; no EC 4.2.3.3 / `mgsA` hit.

Uncertainty statement. `gloA` carries direct *P. putida* functional evidence (strong transfer to KT2440). `gloB`, `yeaE`, `dkgB`, PP_0772 and the DJ-1 candidates are homology-based within KT2440; glyoxalase III and reductive-route assignments are inferred from related organisms and remain to be verified experimentally in KT2440. The absence of `mgsA` is a negative database result and should be confirmed by manual synteny/HMM search.

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